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SDC Hackathon

Technical Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

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SDC HACKATHON 2026

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1. Team & Project Overview

Team Name	EyeScream
Project / Solution Name	BridgeNext
Name (Leader)	Yap Li Shan
Email & Phone (Leader)	lishan0311@gmail.com · 016-272 9309
University Affiliation(s)	Asia Pacific University of Technology & Innovation (APU)
Live Prototype URL	https://bridge-next.replit.app/
Code Repository URL (public / access granted to judges)	https://github.com/MinHueyChan/BridgeNext (public)

2. Architecture & Technology Stack

2.1 System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

BridgeNext runs as one Replit deployment. A React 19 and TypeScript interface, built with Vite, collects the user's occupation, tools, tasks, self-rated skills, interests and weekly study time, or reads a CV in the browser with pdf.js and Tesseract OCR; files are never uploaded. Confirmed profile data goes over HTTPS to a FastAPI backend on the same origin. FastAPI validates every request with Pydantic and runs a deterministic matching engine that scores five career directions and lists skill gaps; the same rules in the browser build a time-budgeted learning pathway with SDG links.

Free-text descriptions use catalogue keyword matching; users review suggestions. Optional Gemini integration through Replit AI Integrations is prepared, with live activation deferred to Phase 2. A signal-layer module discovers Anthropic Economic Index releases on Hugging Face, validates them and writes dated task observations and an audit record to SQLite, keeping the last valid evidence if a check fails. SQLite also stores the reviewed catalogue.

FastAPI serves both the built React files and the JSON API on Replit Deployments. Code is on public GitHub, and credentials stay in Replit Secrets. Appendix A shows the block diagram.

2.2 Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
TypeScript 7.0.2	Frontend language and type checking	Apache-2.0
React 19.3.1 and React DOM 19.3.1	User interface: questionnaire, results, radar and pathway pages	MIT
Vite 8.3.5 and @vitejs/plugin-react 6.1.2	Development server and production build of the frontend	MIT
Recharts 3.10.1	Five-group skill chart	MIT
Lucide React 0.468.0	Interface icons	ISC
pdfjs-dist 6.4.299 (Mozilla PDF.js)	Reads text from PDF CVs in the browser	Apache-2.0

Name & Version	Purpose in Project	License
Tesseract.js 7.0.0 and tesseract.js-core 7.0.0	In-browser OCR for scanned CVs and images	Apache-2.0
Python 3.12	Backend language	PSF-2.0
FastAPI 0.159.1, Uvicorn 0.54.0, Pydantic 2.14.0	JSON API and static hosting, ASGI server, request validation	MIT; BSD-3-Clause; MIT
google-genai 2.32.0	Optional Python client for Gemini through Replit AI Integrations; live activation deferred to Phase 2	Apache-2.0
SQLite 3 (Python sqlite3)	Embedded database	Public domain
requests 2.34.2, anyio 4.15.1, certifi 2026.10.23, idna 3.20	HTTP, async and TLS support dependencies	Apache-2.0; MIT; MPL-2.0; BSD-3-Clause
Node.js 24 with npm; uv	Build tooling on Replit	MIT; MIT or Apache-2.0
DM Sans and Manrope (Google Fonts)	Typography	SIL OFL 1.1
Playwright	Automated browser checks during development (not deployed)	Apache-2.0

3. APIs & Cloud Services

3.1 External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API & Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
Google Gemini API (gemini-2.5-flash), via Replit AI Integrations	Suggests catalogue skill IDs from a free-text work description; output is validated before the user confirms it (Planned in Phase 2)	Sponsor-provided: billed to Replit Core credits; no team-held API key
Hugging Face Hub API and file download (Anthropic Economic Index dataset)	Discovers new releases and downloads the task-level AI-usage CSV, pinned to an immutable revision	Free (public dataset)
jsDelivr CDN	Delivers the tesseract.js worker, core and English language data to the browser the first time OCR runs	Free (public CDN)
Google Fonts API	Loads the DM Sans and Manrope web fonts	Free

3.2 Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
Replit Workspace (Replit)	Development, preview and testing of the full stack	Replit Core (sponsor-provided)
Replit Deployments, Autoscale (Replit)	Hosts the live prototype at https://bridge-next.replit.app/: one FastAPI service serves the React build and the JSON API, with the SQLite file on the same instance	Replit Core credits; Replit-managed region
Replit AI Integrations (Replit)	Managed access to Gemini 2.5 Flash without a team-held API key (Optional Gemini integration prepared for Phase 2; not activated in the preliminary demonstration)	Billed to Replit Core credits
Replit Secrets (Replit)	Keeps the AI integration credentials server-side (Phase 2)	Included in Replit Core
GitHub (GitHub, Inc.)	Public source repository under Apache-2.0	Free plan
RunPod	Not used: the prototype has no GPU workload	—

4. AI Tools, Models & Datasets

4.1 AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
Gemini 2.5 Flash (gemini-2.5-flash)	Google, via Replit AI Integrations	Pre-trained, called through the API with JSON output at temperature 0; no fine-tuning (Phase 2)	Google Gemini API terms; Replit terms of service
Tesseract OCR English model (tessdata 4.0.0), run by tesseract.js 7.0.0	Tesseract OCR project; files served by jsDelivr	Pre-trained, runs in the user's browser	Apache-2.0

4.2 Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
Anthropic Economic Index, release 2026-06-26 (global task-level AI usage, April–May 2026; 6 task statements used)	huggingface.co/datasets/Anthropic/EconomicIndex	Public	CC BY (data); MIT (code)	N
O*NET OnLine occupation and task references (5 occupations)	onetonline.org	Public	CC BY 4.0 (attributed)	N
U.S. BLS Table 1.2, occupational projections 2025–35 (context only)	bls.gov/emp/tables	Public	Public domain (U.S. Government)	N
UN SDG goals and targets	sdgs.un.org	Public	UN terms of use	N
TalentCorp MyCOL 2024/2025; OpenDOSM	talentcorp.com.my/mycol; open.dosm.gov.my	Public (linked only, not processed)	© TalentCorp; © DOSM	N

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
BridgeNext catalogue: 5 directions, 18 skills, 6 task groups, 8 learning resources, SDG and prerequisite mappings	data/catalogue.json in the repository	Self-collected (curated from the sources above and public course pages)	Apache-2.0 (team); source terms apply	N
Synthetic CV sample, personas and test fixtures	Created by the team	Synthetic	Apache-2.0	N

4.3 AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☒ Yes ☐ No

If yes, list them below:

Tool & Version	Used For	Extent of Use (minor / moderate / major)
ChatGPT / OpenAI Codex (OpenAI), October 2026	Writing, refactoring and testing prototype code	Major
Claude (Anthropic), October 2026	Reading competition documents; reviewing the topic, PRD and prototype; drafting the README	Moderate overal1
Replit Agent (Replit), October 2026	Configuring the Replit run workflow and deployment	Minor
OpenAI image generation (OpenAI), October 2026	Generating the chick mascot character sheet from a text brief	Minor

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

A team member reviewed every AI suggestion before use, and AI-drafted text was edited and approved by the team before submission. Code was run locally and on Replit, and 28 automated backend checks,

40 browser–server parity cases and Playwright browser checks cover matching, evidence refresh, input validation and failure handling. We checked statistics, links and licences against their original sources. Results come from traceable rule-based code. Live Gemini activation is deferred to Phase 2; prepared safeguards validate suggested catalogue IDs and require user confirmation.

5. Sponsor-Provided Resources

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

☒ Yes ☐ No

If yes, list them below:

Sponsor	**Resource (API / credits / platform / data)**	**How It Was Used in the Project**
Replit	Replit Core: workspace, Deployments, Secrets and AI credits	Development, hosting the live prototype at bridge-next.replit.app

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

Replit provides the prototype's development, preview and deployment environment. Technically, BridgeNext could run on another compatible host because it uses React, FastAPI and SQLite. Matching, scenarios and pathways do not depend on the model, so the app keeps working with a labelled keyword fallback. RunPod credit was not used because there is no GPU workload.

6. Security, Privacy & Responsible Use

6.1 Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

BridgeNext does not require names, contact details or identity documents, and the presentation interface has no sign-up. Answers, saved pathways and badges remain in browser session storage until the session ends; interface language remains in local storage. Confirmed profiles, including occupation, experience, weekly learning time, interests, task and skill IDs and confidence ratings, are sent over HTTPS to FastAPI for matching but are not persisted by that endpoint. CV files are processed in the browser and never uploaded. "Find matches" uses browser-based catalogue keyword matching. Descriptions are not sent to Gemini; live integration is deferred to Phase 2. SQLite stores catalogue data and evidence history. Separate account-storage APIs exist but are not connected to the presentation interface. API credentials belong in Replit Secrets. Development and testing used only public or synthetic data, with no real CVs.

6.2 Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

Phase 2 model risks: Gemini may misinterpret descriptions or generate unsupported explanations. Prepared safeguards validate catalogue skill IDs, require user review and keep scoring and resource selection deterministic. Live model behaviour remains to be validated.

Preliminary independence: The questionnaire, keyword matching, career comparison and learning pathways work without Gemini. Live activation is deferred to Phase 2.

Data representativeness: Economic Index observations describe sampled global Claude conversations, not Malaysian adoption or job loss; O*NET and BLS are U.S. sources. The app labels each caveat and limits AI evidence to 10% of the score.

Source changes: new releases are validated, and the last valid evidence is kept if a check fails.

Bias and scope: scores use self-assessed confidence and never use age, gender or ethnicity. The catalogue covers five finance and operations directions, and OCR reads English only.

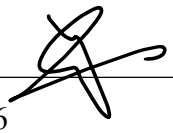
7. Compliance & Readiness Checklist

Tick all boxes that apply before submitting.

- ☒ All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- ☒ All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- ☒ No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- ☒ Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- ☒ Any use of official SDC sponsor resources has been declared in Section 5.
- ☒ The prototype described here is accessible to judges through the links provided in Section 1.
- ☒ All fields in this template have been completed within the word limits stated.

8. Declaration

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.

Signature:  Yap Li Shan (Team Leader)

Date: 24 October 2026